

# Assignment 4: Finite Element Computation in One and Two Dimensions

ME/BME 736 — Advanced Finite Element Methods

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## Submission

Each problem has its own notebook in [assignments/assignment-04/notebooks/](#), which gives the detailed tasks. Set up the environment by following [environment/README.md](#) and use the kernel **Python (Finite Elements)**. Submit the two completed notebooks, renamed `lastname_assignment04_1d.ipynb` and `lastname_assignment04_2d.ipynb`. Each must run without an error after **Restart Kernel and Run All Cells**. No separate report is needed.

## Problem 1 (60 marks): A linear finite element code for a one-dimensional bar

Notebook: `problem1.ipynb`. Reading: Chapter 5.

## Problem 2 (40 marks): Mixed-boundary diffusion with FEniCSx

Notebook: `problem2.ipynb`.

Reading: Chapter 6 and the lecture notebook `notebooks/diffusion/diffusion_2d_demo.ipynb`.